# Predictions: Theory and Practice

Part II: Subjective judgement

Patrick Schmidt
University of Heidelberg

# Introduction

- Judgmental prediction
  - human experts using their knowledge, experience, and intuition
  - often used in situations with limited data or high uncertainty
  - mostly focused on events
- Human judgment can incorporate qualitative factors and adapt to changing circumstances
  - update: So can modern language models
- This part of the lecture will also cover related topics

## Outlook

Part 2 will cover

- 2.1 Elicitation (measurement) of predictions
  - including incentives
- 2.2 Evaluation/analysis of predictions
  - using properties of optimal predictions (Slides 1.2)
- 2.3 Empirical findings: Heuristics and Biases and Superforecasters
- 2.4 Economic Expectations
- 2.5 Market-based predictions

# Elicitation

## Intro

- elicitation = measurement of predictions
  - we say we “elicit” a prediction
  - the word is mainly used for asking or incentivizing individuals
- We cover different approaches
  - humans without incentives
  - humans with incentives
  - markets
- Data and algorithms will follow in Part III

## Goals

- learn how to elicit predictions
  - from experts and non-specialists
  - with incentives and without
  - point and probabilistic predictions

## Background

- This part of the lecture is partly informed by
  - empirical facts from the expert judgement literature
  - game-theoretic insights on incentives
- Context: Experts vs laypersons
  - Often applied for subject-matter experts
  - Not all of those are experts in probability or statistics
  - in research and public policy also applied to representative samples (inflation expectations, KOF business monitor,...)

# Targets

## Target variable

- Target variable needs to be clearly defined
  - note: This is uncommon in daily communication and even businesses
  - ambigous statements are the norm
  - not cleary defined: “The US will set tarifs on trade.”
  - clearly defined: “Average trade tariffs (weighted by net worth) on December 31 2025 will be above 10%.”
- Standard targets are
  - an event
  - a real-valued outcome

## Target property for events

- for an event prediction
  - we normally want to know the probability $\hat{p}$ of the event
  - but could also ask for point prediction: yes/no
  - or for ranges of probabilities (e.g., “30-60%”)
- we have argued that **probabilities** can be connected to decision making consistently

- point predictions do not allow to weigh the utility differences
  - in some situations the risk of being wrong is higher in one direction
- ranges of probabilities normally do not identify a best decision
  - but decision-making with **imprecise probabilities** is also a developped field
  - called decision making under **ambiguity**
  - e.g., maxmin preferences:

$$a^\star([p_l, p_h]) = argmax_a \min_{p \in [p_l, p_h]} E_p[u(a, y)]$$

## Target property for reals

- for real-valued targets
  - ideally we want a probability distribution
  - identified by cumulative distribution function (cdf)
  - or by probability density function (pdf)
  - requires distributional assumptions
  - or approximations (interval probabilities or quantiles)

- As probability distributions for real-valued targets are complicated, we often use point predictions
- How does a point prediction for a real-valued target look like?
  - "My life expectancy is 85 years."
  - "Most likely I will die with 85."
  - $\hat{y} = x'\hat{\beta}_{OLS} = 85$ years
  - I will not live more than 85 years.
- All those statements can be connected to a probability distribution

- Point predictions for real-valued targets should be understood as properties of a probabilistic predictions
  - Mean: "My life expectancy is 85 years."
  - Mode: "Most likely I will die with 85."
  - Mean (will learn later why): $\hat{y} = x'\hat{\beta}_{OLS} = 85$ years
  - Maximum/Minimum: I will not live more than 85 years.
- More common are unspecified point predictions:
  - Unspecified point prediction: "I will die with 85."

- other typical properties of a distribution $P$
  - quantile $q_\alpha$ at level $\alpha$:

$$q_\alpha(P) = \{c | P(y \leq c) = \alpha\}$$

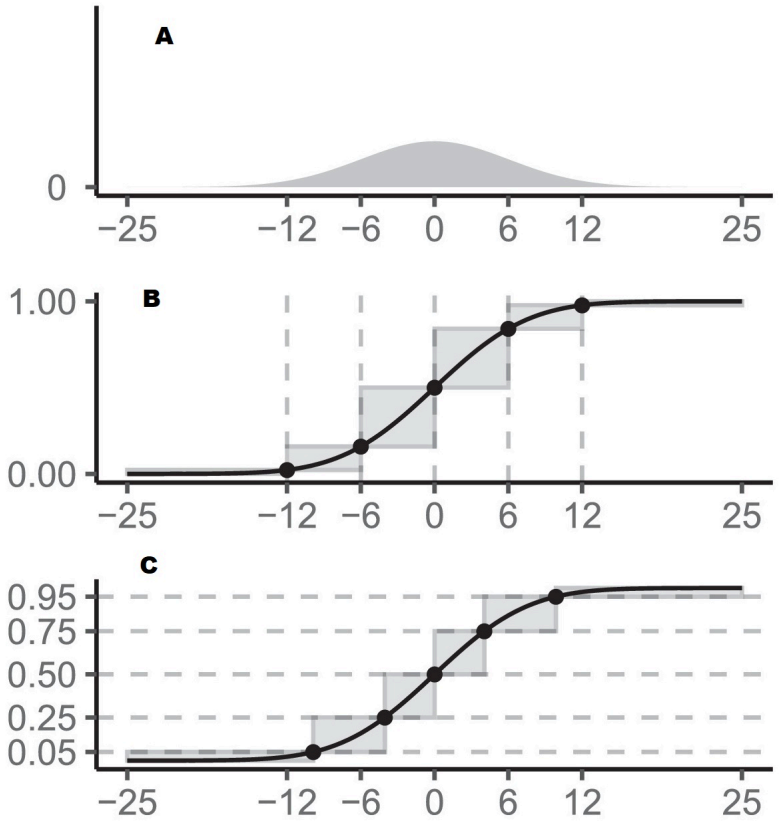
  - interval probability $IP_c$ at point $c$:

$$IP_c(P) = P(y \leq c)$$

  - $\alpha$-confidence intervals $CI_\alpha$:

$$CI_\alpha(P) = \{Interval | P(Interval) = \alpha\}$$

# Approximating prob. distribitions




**Identifying a distribution with interval probabilities and quantiles.** Panel A: pdf, Panel B: cdf identified by interval probabilities, Panel C: cdf identified by quantiles. The gray are illustrates the bounds on the cdf.

# Questions

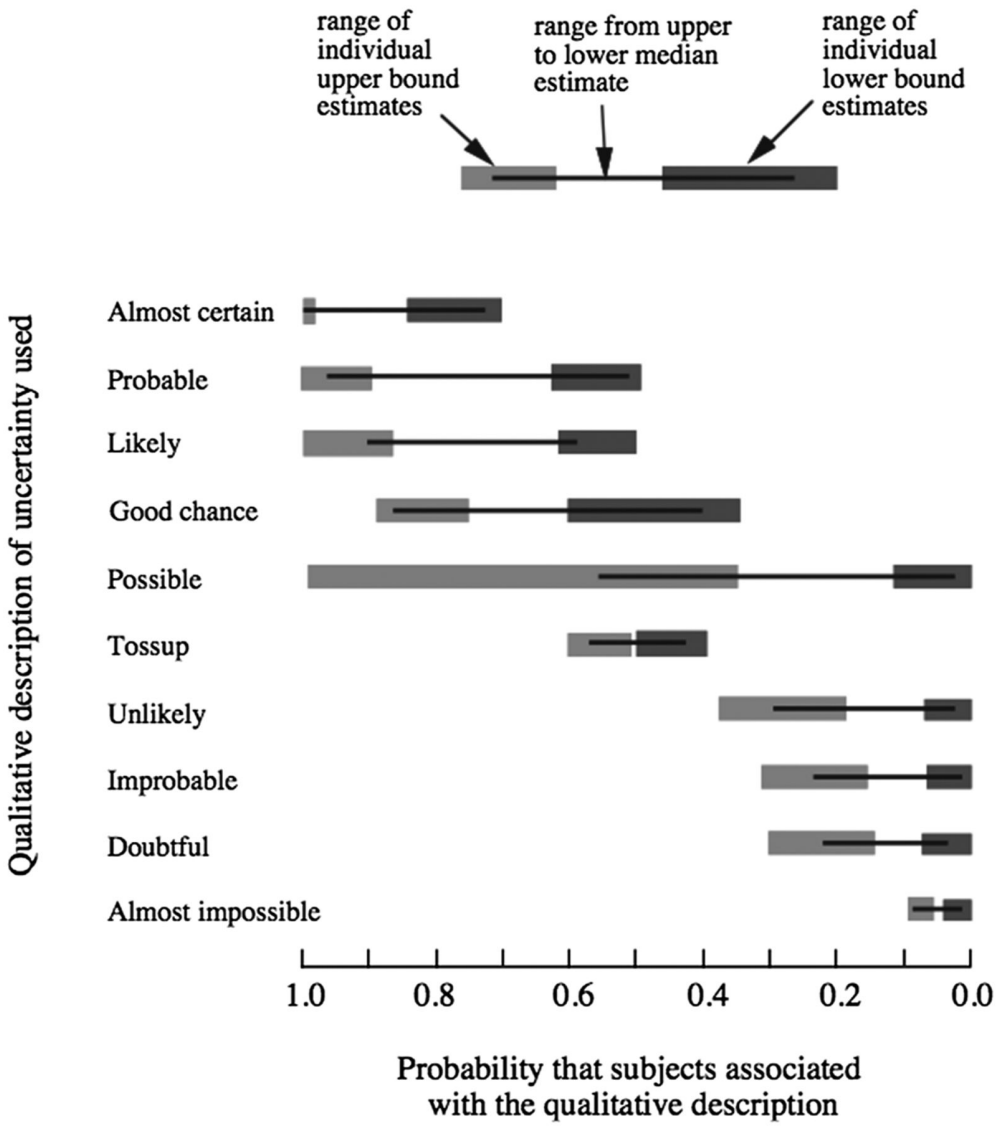
## Things to consider

- information content of answer
  - broadly: range of probabilities $>$ probability $>$ point pred
  - broadly: specific target property $>$ unspecified target
- ease of use
  - broadly: range of probabilities $<$ probability $<$ point pred.
  - often connected to accuracy of report
  - What is better: A bad probability prediction or a good point prediction?

- bias and incentives that question generates
  - suggestive questions
  - framing can change answers
- heterogeneity in target population
  - e.g., research in natural sciences more open to commit to probability predictions than public health experts (Morgan 2014)

## Events

- Probability
  - *What do you think is the probability/likelihood that event happens?*
  - Alternatively communicated as frequency: How often out of 100 times would you expect the event to happen?
- Point prediction
  - *What is more likely? The event or the complement?*
  - less informative than probability
- unspecified scales
  - *How likely is the event? (likert-type scale: very likely, likely, unlikely, very unlikely)*
  - less informative than probability and vague
  - what is the probability associated with “likely”?!




What is the meaning of non-quantitative scales to different people? (Morgan 2014)

## Real-valued targets

- Point prediction:
  - *What do you expect y will be?* (Mean)
  - *What do you think y will be most likely?* (Mode)
  - *What do you think is the highest possible value for y?* (Maximum)
  - generally hard to elicit other properties (e.g., quantiles or variances) without statistical training

- probability predictions with parametric assumptions
  - also requires statistical training
  - even then hard to communicate
- instead:
  - elicit interval probabilities and fit distribution to them
  - Questions: *What is the probability that y is below c?*
  - e.g. …that GDP growth is below 2%?

# Procedures

## 4 steps (Garthwaite et al. 2005)

- setup
  - who is the expert?
  - what is the appropriate target?
  - train expert in elicitation
- elicit
- fit distribution
  - fit a probability distribution to measured properties
- checks
  - Adequate? (If no, go back to elicit)

## Who is the expert?

- different experts may have different opinions
- sometimes subject-matter experts are bad at statistics
  - a doctor in medicine
  - a business leader
- experts have specific biases
  - an expert in public health probably thinks positively about public health interventions
  - an AI entrepreneur probably is optimistic about the technology

## Checks

- Are reports and probability distribution internally consistent?
  - partitions should add up to 100%
  - if $E_1 \subset E_2$, then $P(E_1) \leq P(E_2)$
- Generate statements from probabilities and check with expert
  - "Is global temperature in 2030 more likely to rise above 4 degrees than to stay below 1.5 degrees compared to pre-industrial temperatures?"
  - "Do you expect a rise above 5 degrees to happen in 5 out of 100 cases?"
  - Does the temperature stay in the interval $[2, 4]$ with likelihood of 50%?

## Strategic thinking process

- one can teach the subject-matter expert strategies that help to form better predictions
  - avoid biases
  - improve information content
- we will discuss such strategies at the end of part II (“superforcasting”)
- alternatively, one can take the prediction and do post-processing without the expert
  - standard for models (e.g., in weather forecasting)
  - recalibration (make uncalibrated prediction calibrated)
  - forecast combination (combine multiple predictions to a single, better, one)

## Confidence, Second Order Uncertainty

- generally a probability distribution reflects uncertainty
  - and we argued that it can/should support decisions under uncertainty
- however, subjectively, experts often feel different degrees of confidence about the same probability
- one famous example for second-order uncertainty is from the Intergovernmental Panel on Climate Change (IPCC)
  - used several frameworks over the years to express confidence
  - switched to probabilistic framing with five qualifiers: very low, low, medium, high, and very high
  - example: “In the Northern Hemisphere, 1983–2012 was likely the warmest 30-y period of the last 1,400 years (**medium confidence**).”

- confidence / second-order uncertainty will also be key in model-based algorithmic forecasting (part III)
  - related to “model uncertainty” and “miss-specification”
- for example a linear regression prediction:
  - the model gives a prediction $\hat{y}$
  - model also provides confidence intervals for $y$ (aka “parameter uncertainty”)
  - but if the model is misspecified this uncertainty quantification can be wrong (“model uncertainty”)

## Wisdom of the crowds

- If there are multiple experts, one could elicit multiple predictions
  - How to choose or construct final prediction?
- Wisdom of the crowds: Empirically, we often observe that the **average of all predictions** outperforms most individual predictions
- Important caveats:
  - not always true (small crowd of superforecasters outperforms general crowd)
  - if performance of predictions is known, can be better to choose good performing predictions

## Summary: Elicitation

- Judgement predictions can be elicited from
  - human experts
  - human laypersons (“experts” on public opinion or individual-specific matters)
- elicitation mechanisms differ in
  - information content about uncertainty
  - ease of use
- Next we look at
  - incentives in predictions
  - predicitions gathered from market prices

# Incentives

## Scoring rules

- so far we have taken expert opinion as given
  - expert may have the option to gather information and improve prediction
  - expert may not report truthfully
- Solution: good incentives.
  - what payment structure is a good idea?

## Payment with proper scoring rules

- Consider the following contract structure
  - expert issues prediciton $\hat{p}$
  - target outcome $y$ realizes
  - expert receives bonus $s(\hat{p}, y)$
- What is a good choice for the payment function $s$?
  - proper scoring rules!
- If
  - expert **maximizes expected payout** and
  - $s$ is a **proper scoring rule**,

  then the expert issues their true belief $p$.

## Advantage of proper incentives

- **Truthful** reporting is incentivized
- Incentives reward the inclusion of additional information
  - Including more information increases the expected score for ideal predictions (Holzmann and Eulert 2014)
  - Formally: For two ideal predictions $\hat{p}_1 = E[y|x_1]$ and $\hat{p}_2 = E[y|x_1, x_2]$ it holds that

$$S(\hat{p}_1) \leq S(\hat{p}_2)$$

  - Thus, expert has incentives to include more information in prediction

## The problems with scoring rules

- the properties above hold only under narrow circumstances
- Requirement: Outcomes need to be observable
- Problem: risk aversion
  - if expert is risk averse, she is not maximizing expected payout, but expected utility from it

- Problem: personal stakes
  - if expert has personal stakes (e.g., hedge fund manager) can hedge with prediction (i.e., bias towards unfavorable outcomes)
- Problem: reputation
  - if predictive performance is public, expert has incentives to maximize reputation
  - can lead, for example, to overly extreme or confident predictions
  - media landscape favors extrem statements
  - public institutions (like professors) favor non-commited messaging

# References

Garthwaite, Paul H, Joseph B Kadane, and Anthony O'Hagan. 2005. "Statistical Methods for Eliciting Probability Distributions." *Journal of the American Statistical Association* 100 (470): 680–701.

Holzmann, Hajo, and Matthias Eulert. 2014. *The Role of the Information Set for Forecasting—with Applications to Risk Management*.

Morgan, M Granger. 2014. "Use (and Abuse) of Expert Elicitation in Support of Decision Making for Public Policy." *Proceedings of the National Academy of Sciences* 111 (20): 7176–84.